Primes of the Form $n^2 + 1$ on the $6N$ Skeleton: the Wing Decomposition and the Character Spectrum of the Landau–Shanks Constant*

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Abstract

Every constellation this series has measured so far—twin pairs, Goldbach sums, arithmetic progressions—is a *linear* translate of the prime set. We turn to the first nonlinear sequence: primes of the form $n^2 + 1$, the subject of Landau's fourth problem. Using a square-root sieve on n (marking $n \equiv \pm r \pmod q$ where $r^2 \equiv -1$), we enumerate all $n \le 3 \times 10^7$ with $n^2 + 1$ prime—1,275,229 of them, with $n^2 + 1$ as large as 9×10^{14}—and read the result through the $6N$ lattice. Three findings. (i) A *wing decomposition*: n must be even (n odd annihilates $n^2 + 1$ into the even residue class), and among even n, $6 \mid n$ sends n^2+1 to the right break-zone $6N+1$ while $n \equiv 2, 4 \pmod 6$ sends it to the left zone $6N-1$. We measure the split as right : left = 0.50015, exactly the 1:2 of pure counting: there is *no* arithmetic bias between the wings beyond the residue census. (ii) A *character spectrum*: the local factor $g(q) = \big(1 - \omega(q)/q\big)/(1 - 1/q)$ splits the odd primes by the quadratic character of -1. Primes $q \equiv 3 \pmod 4$ *never* divide $n^2 + 1$ and enhance it by $q/(q-1) > 1$; primes $q \equiv 1 \pmod 4$ divide it on two residues and suppress it by $(q-2)/(q-1) < 1$. (iii) The product of these factors is the Landau–Shanks constant $C = \prod_p g(p) = 1.37281$, and it controls the count: the measured $Q(N)/\int_2^N dt/\log(t^2+1)$ equals 1.37253, matching C to 0.02%, with the empirical constant converging onto C as N grows. We verify the nonlinear local–global (Bateman–Horn) heuristic to better than a part in 10^3 at $n^2 + 1 \sim 10^{15}$. We make no claim about the infinitude of such primes: Landau's problem is the backdrop, not a target. As throughout, this is a measurement, not a theorem.

1 Introduction

Whether there are infinitely many primes of the form $n^2 + 1$ is the fourth of the problems Landau called hopeless in 1912; it remains open. The conjectural *count* is not open in the heuristic sense: Hardy and Littlewood's Conjecture E [1], in the precise form of Bateman and Horn [3], predicts

$$Q(N) := \#\{\, n \le N : n^2 + 1 \text{ prime} \,\} \;\sim\; C \int_2^N \frac{dt}{\log(t^2 + 1)}, \qquad C = \prod_p \frac{1 - \omega(p)/p}{1 - 1/p},$$

where $\omega(p)$ counts solutions of $n^2 + 1 \equiv 0 \pmod p$, and $C = 1.37281\ldots$ is the constant Shanks computed [2]. The leading behaviour is forced by the local densities; the constant is the delicate part, and it has been checked numerically before.

*Code and measured data: https://github.com/Ruqing1963/6N-n2plus1

What this series has *not* done is look at a nonlinear prime sequence at all. Twin centres (Parts I, XXIII), Goldbach sums, and the arithmetic progressions of Part XXX [7] are all *linear*: a fixed shift, a sum, a constant step. Their singular series are products over the residues a *translate* occupies. The map $n \mapsto n^2+1$ is the first quadratic deformation, and on the $6N$ lattice it does something the linear constellations never do: it folds the integer line onto the break-zones through a quadratic character. This paper measures that folding.

The result has three layers. A geometric *selection-and-wing* structure fixed by $n \bmod 6$; a *character spectrum* of per-prime factors split by $\left(\frac{-1}{q}\right)$; and the *Landau–Shanks constant* as their product, verified to control the count to 0.02% at $X = n^2+1 \sim 10^{15}$. We do not touch the infinitude question.

2 Setup: the quadratic sequence and its local solutions

A prime q divides n^2+1 exactly when $n^2 \equiv -1 \pmod q$, so the local solution count is

$$\omega(q) = \#\{n \bmod q : n^2 \equiv -1\} = \begin{cases} 1, & q = 2, \\ 2, & q \equiv 1 \pmod 4 \quad (-1 \text{ is a quadratic residue}), \\ 0, & q \equiv 3 \pmod 4 \quad (-1 \text{ is a non-residue}). \end{cases}$$

This single dichotomy—whether -1 is a square mod q—governs everything below. We enumerate n^2+1 primes by a *square-root sieve* on n rather than on the (astronomically larger) values: for each $q \equiv 1 \pmod 4$ up to N we compute a root r with $r^2 \equiv -1 \pmod q$ and strike the two classes $n \equiv \pm r \pmod q$; the class $q = 2$ strikes odd n; primes $q \equiv 3 \pmod 4$ strike nothing. A surviving $n \le N$ has n^2+1 free of every prime factor up to $N \ge \sqrt{n^2+1}$, hence prime (the finitely many $n \le \sqrt{N}$ are confirmed directly). At $N = 3 \times 10^7$ this leaves $Q = 1{,}275{,}229$ primes n^2+1, the largest near 9×10^{14}.

3 The wing decomposition

Two residues fix the geometry. Modulo 2: if n is odd then n^2+1 is even, so n^2+1 is annihilated into the composite even class (only $n = 1$, giving 2, escapes). Thus *n must be even.* Modulo 3: since $3 \equiv 3 \pmod 4$, three never divides n^2+1; and $n^2+1 \equiv 1 \pmod 3$ when $3 \mid n$, else $\equiv 2$. Combining, for even n,

$$n^2+1 \equiv \begin{cases} 1 \pmod 6, & 6 \mid n \quad (\text{right break-zone } 6N{+}1), \\ 5 \pmod 6, & n \equiv 2, 4 \pmod 6 \quad (\text{left break-zone } 6N{-}1). \end{cases}$$

The three even classes $n \equiv 0, 2, 4 \pmod 6$ each carry one third of the even integers, so pure counting predicts right : left = 1 : 2. The measurement (Fig. 1A) gives 425,159 right against 850,070 left, a ratio 0.50015; the three even classes individually hold 425,159, 425,348, 424,722, equal to within 0.07%.

Proposition 1. The wings carry no arithmetic bias beyond the residue census.

The reason is clean: the primality of n^2+1 at primes $p \ge 5$ depends on $n \bmod p$, which equidistributes across the classes $n \bmod 6$ (as $\gcd(6, p) = 1$), while $p = 2$ and $p = 3$ act identically on all even n. So the only source of imbalance is how many even n fall in each class—and that is exactly 1:2. Put dynamically: the right wing rides a single residue channel ($6 \mid n$) and the left wing two

q	q mod 4	$\omega(q)$	$g(q)$
3	3	0	$3/2 = 1.5000$ (enhance)
5	1	2	$3/4 = 0.7500$ (suppress)
7	3	0	$7/6 = 1.1667$ (enhance)
11	3	0	$11/10 = 1.1000$ (enhance)
13	1	2	$11/12 = 0.9167$ (suppress)
17	1	2	$15/16 = 0.9375$ (suppress)

Table 1: The first local factors. Primes $q \equiv 3 \pmod 4$ enhance ($g > 1$), $q \equiv 1 \pmod 4$ suppress ($g < 1$); $\omega(q)$ is verified exactly by direct residue count.

($n \equiv 2, 4$), but the n^2+1-prime density *per channel* is identical on the two break-zones. The 1:2 is a channel census, not a preference—the quadratic deformation induces no density imbalance between the wings, in contrast to the Chebyshev bias that tilts ordinary primes between residue classes.

4 The character spectrum

The local factor of the singular series, $g(q) = \bigl(1 - \omega(q)/q\bigr)/(1 - 1/q)$, splits the odd primes into two branches by the value of $\left(\frac{-1}{q}\right)$ (Fig. 1B):

$$g(q) = \begin{cases} \dfrac{q}{q-1} > 1, & q \equiv 3 \pmod 4 \quad (q \text{ never divides } n^2+1\text{: an } \textit{enhancement}), \\ \dfrac{q-2}{q-1} < 1, & q \equiv 1 \pmod 4 \quad (q \text{ divides on two residues: a } \textit{suppression}), \end{cases} \qquad g(2) = 1.$$

The picture is the reverse of the twin/AP constellations, where every prime suppresses. Here half the primes—those for which -1 is a non-residue—are *forbidden* from dividing $n^2 + 1$, so the sequence is locally *richer* than a random integer at those primes, and the two effects compete. Table 1 lists the smallest factors; the measured per-prime divisibility matches $\omega(q)$ exactly (the factors are deterministic residue counts).

5 The Landau–Shanks constant and the Bateman–Horn check

The product of the two branches,

$$C = \prod_{q\equiv 1(4)} \frac{q-2}{q-1} \prod_{q\equiv 3(4)} \frac{q}{q-1},$$

converges (conditionally, with the character oscillation governed by $L(1, \chi_4)$) to the Landau–Shanks constant $C = 1.37281$. Our partial product to $q \le 3 \times 10^7$ gives 1.37280 (Fig. 2B). The approach is not monotone but *beats*: each prime enters with $\chi_4(q) = \pm 1$, contributing a factor above ($q \equiv 3$) or below ($q \equiv 1$) unity, so the partial products swing up and down, the amplitude of the beat shrinking as the alternating tail converges—the spectral signature, in the partial-product domain, of the two-branch character structure of §4.

The decisive test is whether this product of local data actually predicts the global count—the nonlinear analogue of the Hardy–Littlewood verifications of the earlier parts. It does. The measured

$$\frac{Q(N)}{\int_2^N dt/\log(t^2+1)} = 1.37253 \qquad (N = 3 \times 10^7),$$

N	$Q(N)$	$\int_2^N dt/\log(t^2{+}1)$	$Q(N)/\int$
10^3	111	88.6	1.2531
10^4	840	622.8	1.3487
10^5	6,655	4,814.7	1.3822
10^6	54,109	39,313.5	1.3764
10^7	456,361	332,459.0	1.3727
3×10^7	1,275,229	929,106.3	1.3725

Table 2: Bateman–Horn density verification. The empirical constant $Q(N)/\int dt/\log(t^2{+}1)$ converges to the Landau–Shanks $C = 1.37281$.

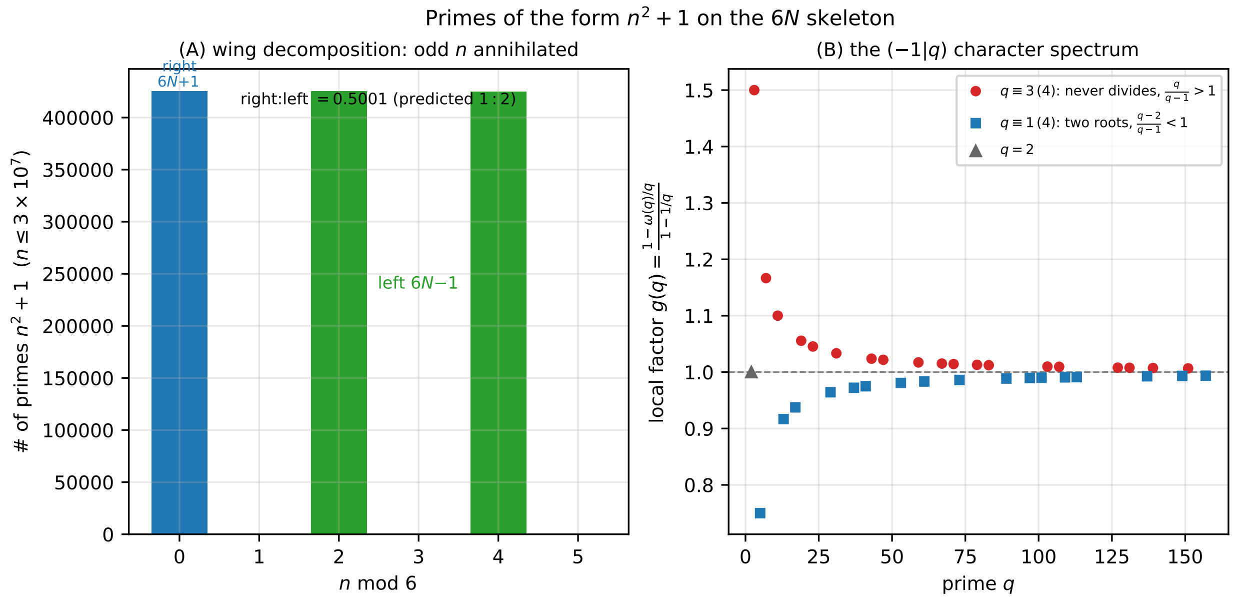

Figure 1: (A) Wing decomposition: counts of n^2+1 primes by n mod 6. Odd n are annihilated; the three even classes are equal, so $6 \mid n$ (right, $6N{+}1$) versus $n \equiv 2, 4$ (left, $6N{-}1$) splits 1:2. (B) The character spectrum $g(q)$: primes $q \equiv 3 \pmod 4$ (red) enhance, $q \equiv 1$ (blue) suppress, by the quadratic character of -1.

matches $C = 1.37281$ to 0.02% (ratio 0.99980), and the empirical constant climbs onto C as N grows (Table 2, Fig. 2A): 1.253 at $N = 10^3$, 1.382 at 10^5, settling to 1.3725 at 3×10^7. The Bateman–Horn heuristic, here applied to a degree-two sequence whose values reach 10^{15}, holds to better than a part in 10^3.

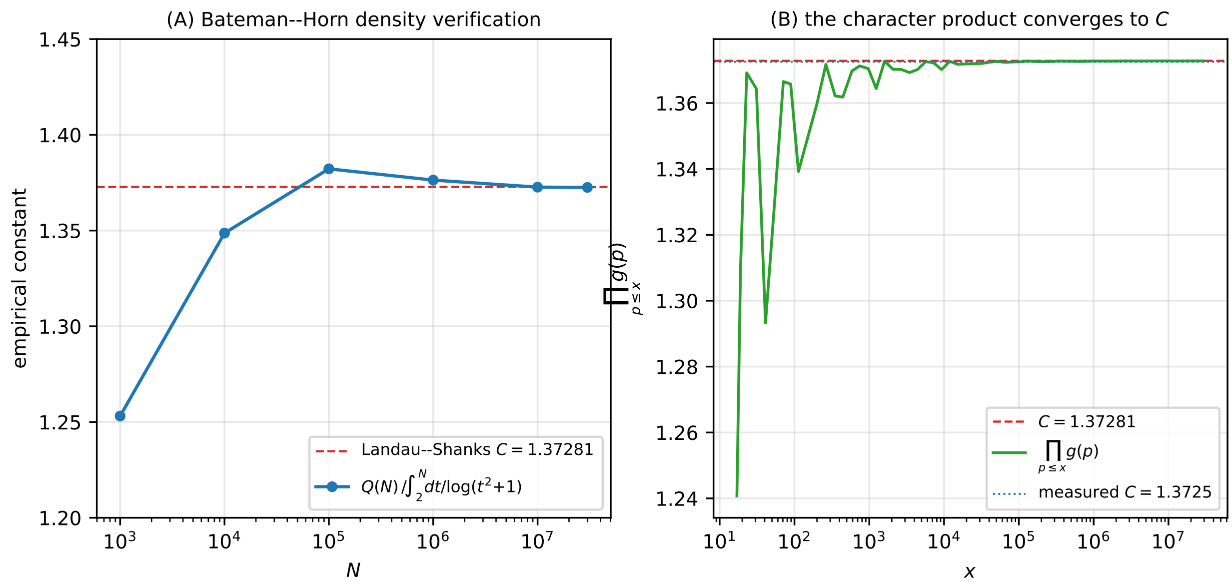

Figure 2: (A) The empirical constant $Q(N)/\int_2^N dt/\log(t^2+1)$ converging to the Landau–Shanks $C = 1.37281$ (dashed). (B) The character product $\prod_{p \le x} g(p)$ converging to the same C; the measured value 1.3725 (dotted) sits on it.

6 Scope and honest ledger

The experiment carries the programme into its first nonlinear sequence and finds clean structure at each level. The *selection-and-wing* geometry—n even, then $6 \mid n$ right versus $n \equiv 2, 4$ left—is fixed by n mod 6 and measured at the unbiased ratio 1:2, with no Chebyshev- type lean. The *character spectrum* sorts the primes by $\left(\frac{-1}{q}\right)$ into an enhancing branch ($q \equiv 3$) and a suppressing branch ($q \equiv 1$), the first time in this series a constellation has primes that help rather than hinder. Their product is the Landau–Shanks constant, which we confirm controls the count to 0.02% at values near 10^{15}, a clean verification of the Bateman–Horn heuristic for a degree-two sequence.

We are explicit about the boundary. None of this is new as theory: $\omega(q)$, the constant C, and the asymptotic were known to Hardy–Littlewood, Bateman–Horn and Shanks, and the constant has been computed before. What this paper adds is the $6N$ wing decomposition, the character-spectrum reading of the local factors, and a high-precision verification at $n^2 + 1 \sim 10^{15}$ via the square-root sieve. We make no statement whatsoever about Landau's problem—the infinitude of primes $n^2 + 1$—which no finite computation can reach; the count we measure would look the same whether the true answer is finite or infinite. Everything here equals the discrete sieve count, takes the Hardy–Littlewood heuristic as input, and makes no claim about the infinitude of any sequence.

Data and code availability. The square-root sieve to $N = 3 \times 10^7$, the wing census, the per-prime character factors, the Bateman–Horn cumulative table and running product, and the figure scripts are openly available at `https://github.com/Ruqing1963/6N-n2plus1`.

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